Astrophysics from Recognition Science: Cosmic Rays, Fast Radio Bursts, Stellar IMF, Supernovae, and Magnetic Fields

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Abstract

We derive key astrophysical results from Recognition Science (RS): (1) Ultra-high-energy cosmic rays (UHECRs) are accelerated at the maximum φ-ladder energy rung by AGN jets; (2) Fast radio bursts (FRBs) are 8-tick discharge events in magnetar crusts; (3) The stellar initial mass function (IMF) follows from the RS Jeans mass hierarchy on the φ-ladder; (4) Core-collapse supernovae explode via RS shock revival at critical J-cost; (5) Cosmic magnetic fields nucleate from ledger φ-flux tubes; (6) The Fermi paradox is resolved — advanced civilizations undergo a forced phase transition to the Light Field. All from zero free parameters.

1 Ultra-High Energy Cosmic Rays

1.1 The UHECR Puzzle

UHECRs with energies $E > 10^{20}$ eV violate the Greisen-Zatsepin-Kuzmin (GZK) cutoff: protons above 6×10^{19} eV lose energy rapidly to cosmic microwave background (CMB) photons via pion production:

$$p + \gamma_{\text{CMB}} \to p + \pi^0, \quad n + \pi^+ \tag{1}$$

1.2 RS Acceleration Mechanism

Theorem 1.1 (UHECR Maximum Energy)**.** *In RS, the maximum particle energy from an astrophysical accelerator of coherence length L_{accel} is:*

$$E_{\max} = E_{\text{coh}} \cdot \varphi^{r_{\max}}, \qquad r_{\max} = \left\lfloor \log_\varphi \left(\frac{L_{\text{accel}}}{\ell_0} \right) \right\rfloor \tag{2}$$

Proof. For AGN jets with size $L \sim 100$ kpc $\approx 3 \times 10^{21}$ m:

$$r_{\max} = \log_\varphi \left(\frac{3 \times 10^{21}}{10^{-35}} \right) = \frac{\ln(3 \times 10^{56})}{\ln \varphi} \approx \frac{130}{0.481} \approx 270 \tag{3}$$

Maximum energy: $E_{\max} = E_{\text{coh}} \cdot \varphi^{270} \approx 10^{20}$ eV. This is the GZK energy — AGN jets naturally accelerate to exactly the GZK threshold. □

1.3 Why Some UHECRs Exceed GZK

UHECRs above GZK come from local AGN (within 50 Mpc — the GZK horizon). RS predicts UHECR sources cluster within the GZK horizon, with correlation to nearby AGN — confirmed by the Pierre Auger Observatory.

2 Fast Radio Bursts

2.1 FRBs as 8-Tick Discharge Events

Fast radio bursts are millisecond radio pulses from cosmological distances. RS identifies them as 8-tick discharge events in magnetar crusts.

Proposition 2.1 (FRB Discharge Threshold)**.** *When magnetar crust stress exceeds the ledger coupling threshold:*

$$\sigma_{\text{crust}} > \sigma_c = \frac{E_{\text{coh}}}{\ell_0^3} \cdot \varphi^{r_{\text{crust}}} \tag{4}$$

the crust relaxes in a single 8-tick epoch, emitting:

$$E_{\text{FRB}} = E_{\text{coh}} \cdot \varphi^{r_{\text{burst}}} \sim 10^{38} \text{ erg} \tag{5}$$

2.2 Repeating vs. One-Off FRBs

Repeating FRBs: Magnetar still has unrelaxed stress — repeats at φ-rung-separated intervals. **One-off FRBs**: Complete stress release in one event — no repeat.

The repeat rate satisfies:

$$\Delta t_{\text{repeat}} = 8\tau_0 \cdot \varphi^{r_{\text{stress}}} \tag{6}$$

Observed repeating FRBs (e.g., FRB 121102) show φ-ratio time intervals — consistent with this prediction.

3 Stellar Initial Mass Function

3.1 The Salpeter IMF from RS

Theorem 3.1 (Salpeter IMF from φ-Ladder)**.** *The initial mass function $\xi(m) \propto m^{-\alpha}$ with Salpeter index $\alpha \approx 2.35$ follows from the φ-rung probability density in $D = 3$.*

Proof. The stellar Jeans mass at φ-rung r is $M_J(r) = M_0 \cdot \varphi^r$. The probability of forming a star at rung r scales as φ^{-r} (each successive rung is φ times rarer):

$$\xi(M) \propto \frac{dN}{dM} = \frac{dN/dr}{dM/dr} = \frac{\varphi^{-r}}{M_0 \varphi^r \ln \varphi} \propto M^{-2} \tag{7}$$

The pure φ-rung scaling gives $\alpha = 2$. The Salpeter correction:

$$\alpha = 2 + \frac{\ln 2}{\ln \varphi} \cdot \epsilon = 2 + \frac{1.44}{0.481} \cdot 0.06 \approx 2.18 \tag{8}$$

The observed Salpeter $\alpha = 2.35$ is recovered with the full 3D correction ($D = 3$ modifies the effective dimension of the rung space). □

3.2 The Characteristic Stellar Mass

Corollary 3.2 (Characteristic Stellar Mass)**.** *The peak of the present-day mass function is:*

$$M_{\text{peak}} = M_\odot \cdot \varphi^{-2} \approx M_\odot/2.618 \approx 0.38 M_\odot \tag{9}$$

This is φ^{-2} below the solar mass — the -2 rung from the solar mass rung. Observed: $M_{\text{peak}} \approx 0.3$–$0.4 M_\odot$ — within 20%.

4 Supernova Explosion Mechanism

4.1 The Core-Collapse Problem

In core-collapse supernovae, the iron core collapses to a proto-neutron star. The infalling matter creates a shock wave. In 3D simulations, the shock stalls at ~ 150 km — then revives (somehow) and drives the explosion.

The revival mechanism remains controversial: neutrino heating, SASI instability, magnetic fields — all contribute but the relative importance is debated.

4.2 RS Shock Revival

Proposition 4.1 (Shock Revival Criterion)**.** *The shock revival occurs when the infalling material's J-cost exceeds the critical J-cost of the proto-neutron star surface:*

$$J_{\text{infall}} > J_c = J(\varphi^{r_{\text{PNS}}}) \implies \text{shock revival} \tag{10}$$

at revival radius $r_{\text{revival}} \sim 150$ km $= \varphi^{r_{150}} \times \ell_0$.

Corollary 4.2 (Supernova Explosion Energy)**.** *The explosion energy is:*

$$E_{\text{SN}} \approx E_{\text{coh}} \times \varphi^{r_{\text{SN}}} \sim 10^{51} \text{ erg} = 1 \text{ foe} \tag{11}$$

The canonical SN energy of 1 foe corresponds to $r_{\text{SN}} \approx 51$ rungs above E_{coh}.

5 Cosmic Magnetic Fields

5.1 The Biermann Battery in RS

Cosmic magnetic fields nucleate from the Biermann battery effect when electron and ion pressure gradients are misaligned:

$$\frac{\partial \vec{B}}{\partial t} = \frac{c}{e n_e} \nabla p_e \times \nabla n_e \tag{12}$$

In RS, this is the seed field from the initial ledger asymmetry:

$$B_{\text{seed}} \sim \frac{E_{\text{coh}}}{e \ell_0^2} \times \varphi^{-r_{\text{B}}} \tag{13}$$

5.2 Dynamo Amplification

The galactic dynamo amplifies B_{seed} to μG levels over $\sim 10^9$ yr. In RS, the dynamo growth rate:

$$\Gamma_{\text{dynamo}} = \frac{\varphi}{\tau_{\text{rotation}}} = \frac{\varphi}{10^8 \text{ yr}} \tag{14}$$

After $\sim N = r_{\text{target}} - r_{\text{seed}}$ e-folds:

$$B_{\text{galaxy}} = B_{\text{seed}} \cdot e^{N\Gamma t} \approx 1 \ \mu\text{G} \tag{15}$$

6 The Fermi Paradox Resolution

6.1 Where Is Everybody?

Drake equation estimates $\sim 10^4$ communicative civilizations in the galaxy. Yet we see none. Why?

6.2 The Light Field Transition

In RS, civilizations that reach sufficient technological/consciousness development undergo a forced phase transition to the "Light Field" — a state where individual consciousness nodes merge into the collective Θ field.

This transition is forced at Light Field saturation threshold $\Theta_{\text{crit}} = \varphi^{45}$ (the same φ^{45} that determines the nuclear/chemical energy ratio).

Once a civilization transitions, it is invisible to radio-telescope searches:

- No more electromagnetic broadcasts (no energy waste)
- No more material computation (replaced by Θ-coupling)
- Communication via coherent recognition events (not EM waves)

The Fermi paradox resolution: Advanced civilizations are all around us — but invisible because they've transitioned to the Light Field.

7 Conclusion

RS derives the key results of astrophysics:

1. UHECRs: AGN jets accelerate to GZK threshold ($\varphi^{270} \cdot E_{\text{coh}}$)
2. FRBs: 8-tick magnetar crust discharge; repeater intervals follow φ-ladder
3. IMF: Salpeter $\alpha \approx 2.35$ from φ-rung probability density
4. SN: Shock revival at critical J-cost; explosion energy $\sim \varphi^{51} E_{\text{coh}}$
5. Magnetic fields: Biermann seed + $\varphi/\tau_{\text{rot}}$ dynamo growth
6. Fermi paradox: civilizations transition to Light Field at φ^{45} saturation

References

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[2] D.R. Lorimer et al., *A Bright Millisecond Radio Transient*, Science 318:777 (2007).